INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

TEST SET GAUGE, VACUUM INDICATING

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID describes a Vacuum Gauge which indicates accurate high vacuum pressure measurement values for calibration, orientation of test measurement and diagnostic equipment and the development, calibration, repair and maintenance of thermal processing, film deposition, spectrographic and an-aerobic chemical processes. This CID lists the minimum performance, design, functional, environmental, reliability requirements and quality assurance provisions for the Vacuum Gauge indicator, herein; referred to as the Vacuum Gauge test set.

2. CLASSIFICATION. The Vacuum Gauge test set covered by this CID will be commercially available equipment that has factory installed options and/or may be modified to the extent necessary to meet the following description. The test set shall be Class 3, in accordance with MIL-PRF-28800.

3. SALIENT CHARACTERISTICS.

3.1 Item Description. The Vacuum Gauge test set is designed to serve as a reference value for calibration of high vacuum applications. The Vacuum Gauge test set shall be capable of measuring vacuums and display the measurements in Torr's, Pascal's and mBar's.

3.2 Vacuum Range Parameters. The Vacuum Gauge test set shall be capable of measuring and displaying scalar vacuum values in the in Pascal's, Torr's and mBar's as a function of the sensor/transducer from 5•10-2 (± 2•10-2) Torr's to 5•10-10 (±1•10-10) Torr's for broad range Ion Gauge Glass transducers.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC 404 SCMS/GUEEA, 235 Byron Street Suite 19A, Robins AFB, GA 31098-1670 or emailed to SPEC99@us.af.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/ .

AMSC N/A FSC 6685

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3.3 Vacuum Scales. The Vacuum Gauge test set shall be minimally capable of displaying measurement values in Linear, Decade, Logarithmic & Lin/Log modes.

3.4 Instrument Interface. The Vacuum Gauge test set shall be capable of displaying a wide range of pressure levels by interfacing with several transducers spanning a pressure range from 103 Torrs to 10-11 Torrs.

3.5 Instrument Expansion Ports or Slots. The Vacuum Gauge test set shall be equipped with a minimum of two (2) expansion ports or slots to perform two simultaneous measurements.

3.6 Instrument Interface Current Capacity. The Vacuum Gauge test set shall be capable of providing current on demand ≥ 125 mA to each sensor or transducer concurrently.

3.7 Instrument High Current Capacity. The Vacuum Gauge test set shall be configured with two (2) or more high current power terminals or connectors and be capable of providing ≥ 2 A to independent circuits by means of relays, either solid state or electromechanical.

3.8 Instrument Output Voltage. The Vacuum Gauge test set shall be capable of providing a proportional analog (linear) voltage output terminals or receptacle/jack with a typical value from 0 to 10 Volts DC into a high impedance (≥ 1kΩ) load.

3.9 Instrument Leak Function. The Vacuum Gauge test set shall have a visual and audible leak rate warning alert.

3.10 Display Information. The Vacuum Gauge test set shall provide a direct readout display and the display accuracy shall be a minimum of one (1) digit.

3.11 Design and Construction. The Vacuum Gauge test set shall be used in both bench top and rack mounted environments. The Vacuum Gauge test set is a commercial off the shelf item subject to commercial products regulatory standards. The Vacuum Gauge test set shall comply with all applicable MIL-PRF-28800 sections. The Vacuum Gauge test set shall be designed and constructed in accordance with all applicable European Union (EU) requirements in order to have the “CE” marking affixed.

3.12 System Safety. The system shall comply with all sections of the Underwriters Laboratories (UL) standard IEC-61010-1 for electrical safety.

3.13 Electrostatic Discharge (ESD). The Vacuum Gauge test set EMI and EMC emissions shall be in accordance with the following radiated emission and susceptibility requirements of MIL-PRF-28800 section 4.5.6.5, MIL-STD-461: RE102 and RS103, IEC-61000-4 thru IEC-61000-6 and emitted radiation absorption and radiation limits under FCC Part 15.109 and 15.209.

3.15 Operating Temperature Range. The Vacuum Gauge test set shall meet the specifications listed herein with full capabilities within the temperature range of 32° to 122°F (0°C to 50°C) after performing tests in MIL-PRF-28800 section 4.5.5.1.1 table 11 for class 3 equipment classification.

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3.16 Storage Temperature Range. The Vacuum Gauge test set shall meet and/or exceed manufacturer's specifications for stored equipment after tests of MIL-PRF-28800 section 4.5.5.1.1 table 11 for temperature range and relative humidity storage for class 3 equipment classification.

3.17 Altitude Requirements. The Vacuum Gauge test set shall meet its operating performance and accuracy while operating at an altitude equal or greater than 5,000 ft. (1,524 meters). The equipment shall meet performance and accuracy requirements, for non-operating storage up to 10,000 (3048 meters).

3.18 Calibration and Maintenance Adjustments. No special tools or software shall be required for alignment or adjustment except those furnished with the Vacuum Gauge test set.

3.19 Calibration Period. The design of the Vacuum Gauge test set shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the instrument internal modules. The calibration interval shall be based on a period of one calendar year or an operating time of 2,080 hours or whichever occurs first.

3.20 Reliability. The design of the Vacuum Gauge test set shall be such that under normal use and operation the mean time between failures (MTBF) exceeds the minimum of 1,500 hours of operation in accordance with MIL-PRF-28800. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with the applicable section of MIL-PRF-28800. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800 sections.

3.21 Maintainability. The Vacuum Gauge test set shall meet the maintainability criteria in accordance with MIL-STD-1472 paragraphs 5.9 through 5.9.18 and 5.9.21 and MIL-PRF-28800 sections 3.14.1 through 3.14.4 and sections 6.6.2.4 and 6.6.2.4.1.

3.22 Preventive Maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800 sections 3.14.2 and 3.14.3.

3.23 Corrective Maintenance. The Vacuum Gauge test set shall be designed to meet reparability criteria in accordance with section 3.14.3 sub sections a through d of MIL-PRF-28800.

3.24 Electrical Power Sources and Connections. The Vacuum Gauge test set shall operate from nominal commercial, military, aircraft and power sources with a nominal voltage of 120 VRMS Single Phase AC (±10%) at line frequencies of 50 Hz & 60 Hz (±5%), in accordance with MIL-PRF-28800 paragraph 3.5.1.12 and 3.5.1.2.2a. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein. AC input power shall be used to power the Vacuum Gauge test set and recharge batteries for nonvolatile memory if equipped with them.

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3.25 Dimensions. The Vacuum Gauge test set shall comply with width and depth of ECIA standard ECA/EIA-310 equipment rack configuration. The width shall be the 19” when configured with rack adapter/mount hardware and include a rack-mount conversion kit if required at organizational level. The height shall be compatible with standard ECIA height units. One unit or "U" is defined by ECA/EIA-310 to be 1.75 inches or 44.45 mm.

3.26 Weight. The Vacuum Gauge test set system shall comply with MIL-STD-1472 section 5.8.6.3.1a for weight and size restrictions for single person lift and handling. The total weight of the Vacuum Gauge test set, excluding accessories and manuals, shall not exceed a one-person lift of 25 pounds, (11.4 kilograms).

4. REGULATORY REQUIREMENTS

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISION

5.1 Verification. Verification of the delivered Vacuum Gauge test set will be performed at the delivery point by the end user.

6. PACKAGING

6.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 7.3). When packaging of materiel is to be performed by Department of Defense (DoD) or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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7. NOTES

7.1 Intended use. The Vacuum Gauge is used in the development, calibration repair, service of test measurement and diagnostic equipment in support of aviation machinery, laboratory work & chemical processes.

7.2 Source of documents.

7.2.1 The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/.

7.2.2 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://quicksearch.dla.,mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg. 4, Section D, Philadelphia, PA 19111-5094. Phone (215) 697-1187, FAX (215) 697-2978, or Customer Service (215) 697-2667/2179.

7.2.3 ECIA standards can be obtained at 1111 Alderman Drive, Suite 400, Alpharetta, GA 30005, phone 678-393-9990 or online at http://www.eciaonline.org/default.aspx .

7.2.4 IEC documents may be obtained at http://www.eciaonline.org/default.aspx or from Electronic Components Industry Association, 111 Alderman Drive, Suite 400, Alpharetta, GA 30005.

7.2.5 FCC standards can be obtained at online at 445 12th Street SW, Washington, DC 20554, 1-888-225-5322 or online at http://www.fcc.gov/ .

7.3 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the CID.
b. Product verification delivery point (see 5.1)
c. Packaging requirements (see 6.1)

7.4 National Stock Number (NSNs). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

NSN
6685-01-542-3030
6685-01-632-3360

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7.5 Key Words.

Calibration repair
High vacuum measurement
High vacuum pressure
Reference value indicator

CONCLUDING MATERIAL

Custodians:
Army - AV
Air Force - 84

Preparing Activity:
Air Force - 84

Reviewer:
Army – CE3
Air Force - 99

Agent:
Air Force – 99
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